SILVER (XAG/USD)

Silver Market Analysis & Price Forecasting

A Decade of Daily Trading Data (2016–2026): Exploratory Analysis, Visualization & Time Series Forecasting

Project Report

Dataset: silver_prices_10y.csv | 2,606 daily trading sessions | Jan 29, 2016 – Jan 23, 2026

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1. Executive Summary

This report presents a comprehensive analysis of daily silver (XAG/USD) price data spanning ten years, from January 29, 2016 to January 23, 2026 — 2,606 trading sessions in total. The dataset combines standard OHLCV fields with pre-computed technical indicators (rolling moving averages, daily returns, and 20-day realized volatility) and calendar decomposition fields, making it suitable for both descriptive market analysis and predictive modeling.

The analysis is organized into three parts: (1) an exploratory data analysis (EDA) covering price trends, return behavior, volatility regimes, seasonality, and inter-feature correlation; (2) formal time series diagnostics, including seasonal decomposition and stationarity testing; and (3) a forecasting exercise comparing a naive baseline, a classical ARIMA model, and a Random Forest regressor on a 60-trading-day held-out test window, followed by a 30-day-ahead forecast.

Key Findings at a Glance

- Silver traded between $12.44 and $35.36 per ounce over the decade, with a mean close of $21.73 and a standard deviation of $5.66 — reflecting several distinct multi-year regimes rather than one steady trend.
- Daily returns average +0.042% with a standard deviation of 1.49%, near-zero skew (0.044), and slightly positive excess kurtosis (0.070) — consistent with a roughly symmetric but mildly fat-tailed return distribution typical of commodities.
- The single worst trading day was -4.93% (January 31, 2017); the single best was +5.76% (November 17, 2016).
- Realized volatility peaked sharply in July 2020, in the aftermath of the COVID-19 market shock, with 20-day volatility exceeding 2.4% — roughly 60% above the full-sample average.
- An Augmented Dickey-Fuller test confirms that price levels are non-stationary ($p = 0.674$) while first differences are stationary ($p < 0.001$), supporting an ARIMA(p,1,q) specification.
- On the 60-day test window, the naive “last value” baseline (MAE $0.43, MAPE 1.29%) was difficult to beat: Random Forest came close (MAE $0.46, MAPE 1.39%) while ARIMA(5,1,0) underperformed both (MAE $1.17, MAPE 3.65%) — a reminder that short-horizon commodity prices behave close to a random walk.
- The 30-trading-day-ahead ARIMA forecast projects a price of approximately $31.52 by March 6, 2026 (95% CI: $28.20–$34.85), essentially flat relative to the last observed close of $31.48 on January 23, 2026.

2. Dataset Overview

The dataset contains 2,606 rows and 16 columns of daily trading data. Each row corresponds to one business-day trading session; weekends and market holidays are excluded, giving an average of approximately 261 sessions per year — consistent with standard trading-calendar conventions.

2.1 Column Structure

Column	Description
Date	Trading date (YYYY-MM-DD)
Open, High, Low, Close	Standard OHLC prices, USD per troy ounce
Adj Close	Close price adjusted for splits/distributions
Volume	Total units traded in the session
Daily_Return	Percentage change vs. previous close (null on row 1)
MA_20 / MA_50 / MA_200	Simple rolling means of Close (20/50/200-day windows)
Volatility_20	20-day rolling standard deviation of Daily_Return
Year, Month, Day_of_Week, Quarter	Calendar decomposition of Date

2.2 Data Quality Checks

Data quality checks confirm the dataset is clean and internally consistent:

- Zero duplicate trading dates.
- Zero OHLC logical inconsistencies (High is always ≥ max(Open, Close, Low); Low is always ≤ min(Open, Close, High)).
- All missing values are fully explained by rolling-window warm-up periods rather than data gaps (table below).

Column	Missing Count	Missing %	Explanation
Daily_Return	1	0.04%	No prior close available for row 1
MA_20 / Volatility_20	19–20	0.73–0.77%	First 20-day window incomplete
MA_50	49	1.88%	First 50-day window incomplete
MA_200	199	7.64%	First 200-day window incomplete

2.3 Descriptive Statistics — Price & Volume

Statistic	Open	High	Low	Close	Volume
Mean	21.73	21.89	21.57	21.73	11,879,300
Std Dev	5.66	5.70	5.62	5.66	7,937,814
Min	12.36	12.49	12.30	12.44	975,449
25th pct	16.09	16.21	15.97	16.10	6,502,228
Median	22.56	22.74	22.42	22.57	9,929,313
75th pct	25.62	25.80	25.40	25.57	14,977,560
Max	35.40	35.73	35.00	35.36	84,993,490

3. Exploratory Data Analysis

3.1 Long-Term Price Trend

Figure 1 traces the closing price across the full decade with 50-day and 200-day moving averages overlaid. The series shows several clearly distinguishable regimes rather than a single trend: a range-bound 2016–2019 period, a sharp COVID-era crash and rebound in 2020, a volatile-but-sideways 2021–

2023 stretch as global interest rates rose, and a strong rally through 2024–2025 that carried silver from the low-$20s to above $30/oz before easing back in early 2026.

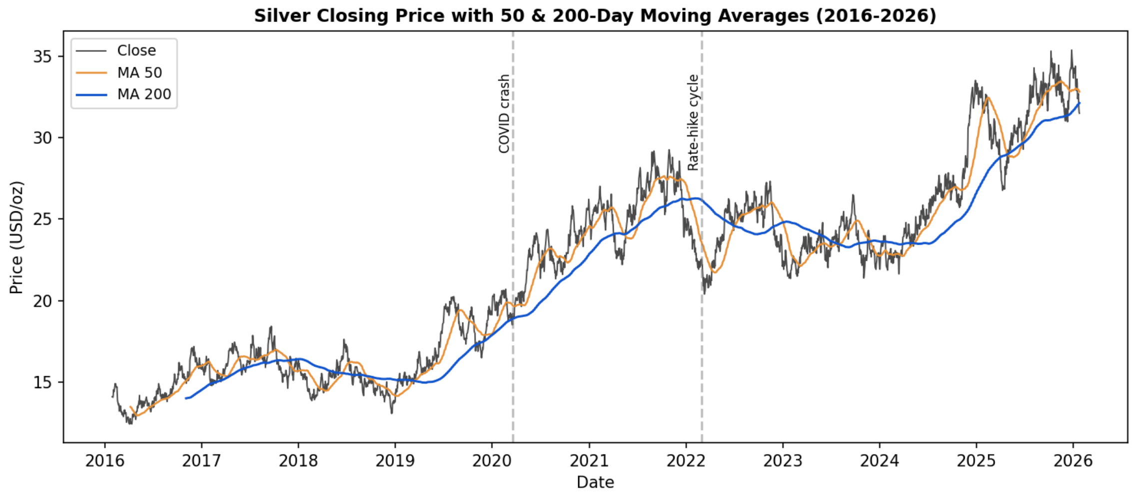

Figure 1. Silver closing price with 50-day and 200-day moving averages, 2016–2026.

Year-over-Year Performance

Year	Open (1st day)	Close (last day)	% Change
2016	14.11	15.83	+12.2%
2017	15.79	15.87	+0.5%
2018	16.34	13.80	-15.6%
2019	14.34	19.13	+33.4%
2020	19.72	25.90	+31.3%
2021	25.39	24.71	-2.7%
2022	24.30	22.10	-9.0%
2023	22.22	22.76	+2.4%
2024	21.94	32.65	+48.8%
2025	32.95	33.69	+2.2%
2026 (partial, through Jan 23)	33.64	31.48	-6.4%

2024 stands out as the strongest year in the sample (+48.8%), while 2018 was the weakest full year (-15.6%). This is consistent with silver's dual identity: 2018's decline coincided with a strengthening US dollar and rising real rates, while 2024's surge aligned with renewed safe-haven demand and accelerating industrial (solar/electronics) consumption.

3.2 Trading Volume

The five highest-volume sessions in the dataset are concentrated around sharp single-day price moves, confirming that volume spikes coincide with volatility events rather than occurring randomly:

Date	Close	Volume	Daily Return
2017-09-22	17.55	84,993,489	-1.96%
2025-02-25	29.78	69,925,981	-2.54%
2017-05-30	15.65	69,079,169	-2.12%
2016-04-14	12.83	60,745,493	+1.84%

Date	Close	Volume	Daily Return
2019-07-24	19.39	60,363,175	+0.68%

3.3 Daily Return Behavior

Figure 2 shows the distribution and time series of daily percentage returns. Returns average +0.042% per day with a standard deviation of 1.49%. The distribution is close to symmetric (skewness = 0.044) with mildly fatter tails than a normal distribution (excess kurtosis = 0.070) — large single-day moves are somewhat more common than a Gaussian model would predict, though the effect is modest over this particular sample.

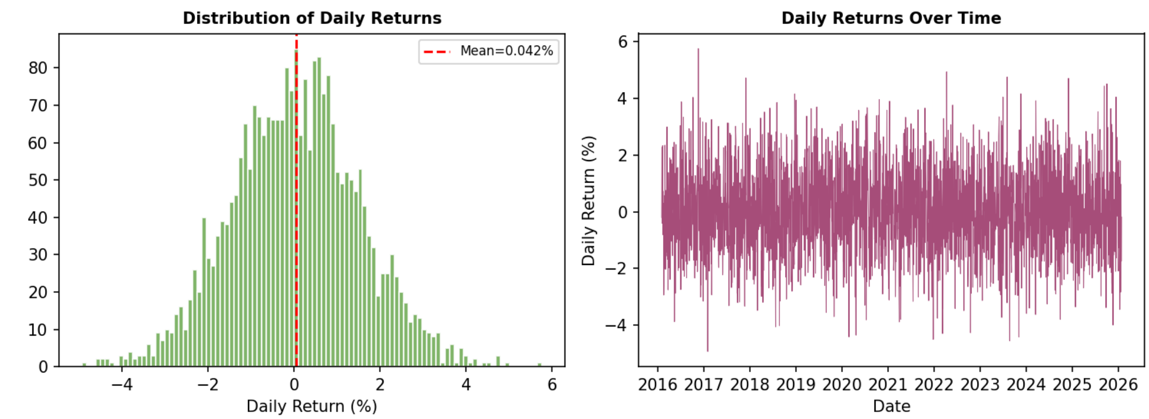

Figure 2. Distribution of daily returns (left) and daily returns over time (right).

The single worst trading day in the sample was -4.93% on January 31, 2017; the single best was +5.76% on November 17, 2016 — both occurring relatively early in the sample window, well before the COVID-era volatility that many would expect to dominate the extremes.

3.4 Volatility Regimes

Figure 3 plots the 20-day rolling volatility of returns, revealing that turbulent periods cluster in time rather than appearing as isolated spikes — a hallmark of financial time series known as volatility clustering. The five most volatile 20-day windows in the entire dataset all fall in July 2020, in the immediate aftermath of the COVID-19 shock, with rolling volatility reaching roughly 2.42% versus a full-sample average of 1.49%.

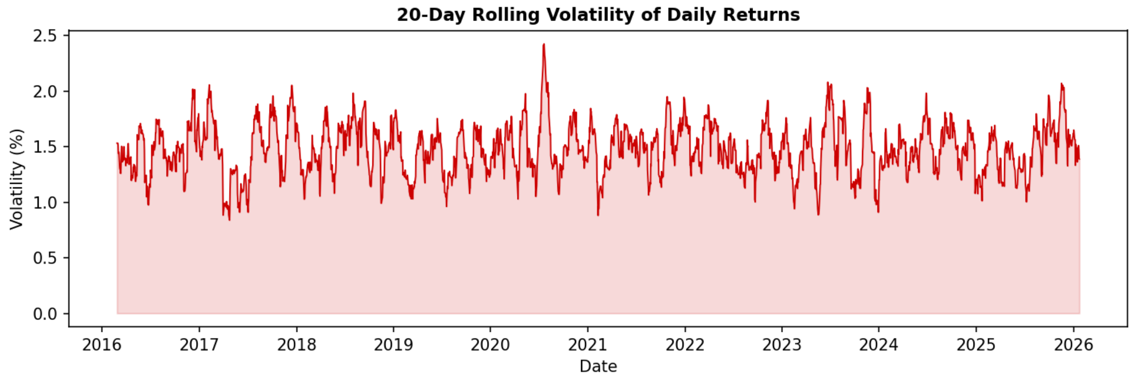

Figure 3. 20-day rolling volatility of daily returns.

3.5 Seasonality

Grouping average daily returns by year highlights meaningful year-to-year variation (from -0.39% in the partial 2026 sample to +0.15% in 2024), while month-of-year and day-of-week breakdowns (see the companion notebook, Section 3.6) show comparatively modest calendar effects relative to overall volatility — consistent with the view that silver's daily moves are driven primarily by macro and industrial-demand news flow rather than a strong recurring seasonal pattern.

3.6 Feature Correlations

Figure 4 shows pairwise correlations across price, volume, return, moving-average, and volatility fields. As expected, Open/High/Low/Close/Adj Close are almost perfectly correlated with one another and with the moving averages (0.94–1.00), since MAs are direct rolling transformations of Close. Volume and Daily_Return show weak correlation with price levels (−0.03 and 0.03 respectively) — price level and short-term trading activity are largely independent in this market. Notably, Volatility_20 correlates only weakly with price level (~0.05) and with Daily_Return itself (~0.01), confirming that volatility is best understood as a separate, regime-driven dimension rather than something that simply tracks price direction or level.

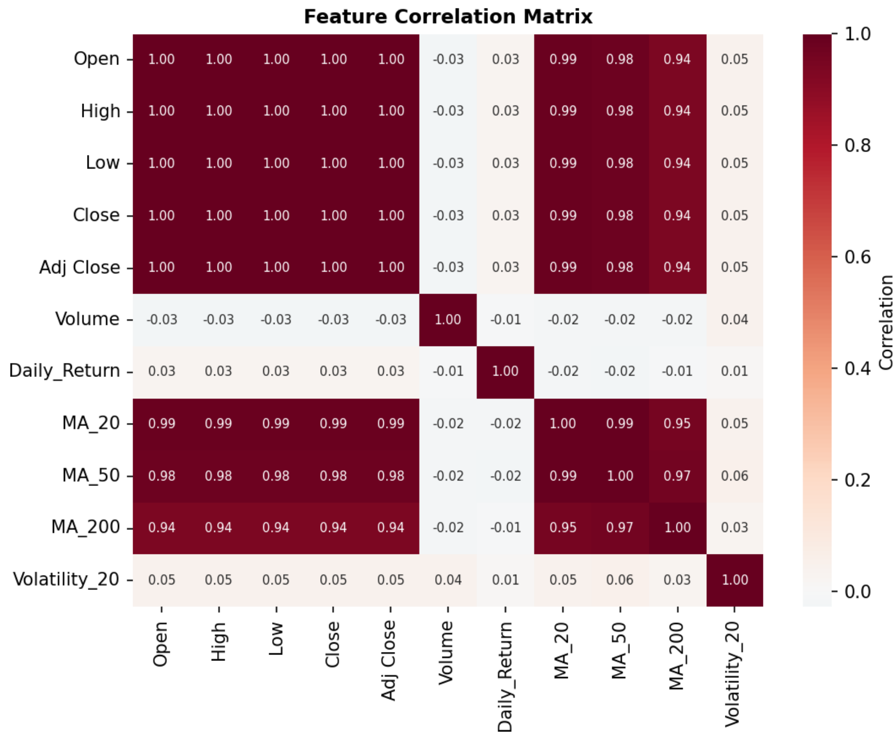

Figure 4. Correlation matrix across price, volume, return, moving-average, and volatility features.

4. Time Series Decomposition & Stationarity

An additive seasonal decomposition (with a 252-trading-day annual period) separates the Close series into trend, seasonal, and residual components, isolating the long-run price trajectory from any recurring annual pattern and short-term noise (see companion notebook, Section 4, for the full decomposition plot).

4.1 Augmented Dickey-Fuller (ADF) Stationarity Tests

Series	ADF Statistic	p-value	1% Critical Value	Conclusion
Close (levels)	-1.198	0.674	-3.433	Non-stationary
Close (first difference)	-27.233	<0.001	-3.433	Stationary

Price levels fail to reject the null hypothesis of a unit root ($p = 0.674$), confirming that silver's price behaves like a non-stationary random-walk-type series, as is typical for commodity and asset prices. The first difference (day-over-day price change) strongly rejects non-stationarity ($p < 0.001$), indicating that a single order of differencing ($d = 1$) is sufficient to stabilize the series — directly informing the ARIMA(p,1,q) specification used in Section 5.

5. Forecasting Models

Three forecasting approaches were built and evaluated on an identical held-out test window of the final 60 trading days (November 3, 2025 – January 23, 2026), with the preceding 2,546 sessions used for training:

- Naive baseline — tomorrow's price is assumed equal to today's actual price.
- ARIMA(5,1,0) — a classical autoregressive model on first-differenced prices, with the order chosen based on the stationarity results in Section 4.
- Random Forest Regressor — trained on engineered features (lagged closes at 1/2/3/5/10 days, MA_20, MA_50, Volatility_20, and Volume).

5.1 Model Performance Comparison

Model	MAE (USD)	RMSE (USD)	MAPE (%)
Naive (last value)	0.4253	0.5388	1.29%
ARIMA(5,1,0)	1.1743	1.4389	3.65%
Random Forest	0.4559	0.5572	1.39%

The naive baseline was the strongest performer by a small margin, with Random Forest close behind and ARIMA a clear third. This result is common in short-horizon financial forecasting: daily commodity prices are close to a random walk, so a persistence forecast is a genuinely hard benchmark to beat, and any model that adds meaningful value typically needs either a longer horizon, exogenous predictors (e.g., gold, DXY, real rates), or an explicit volatility/regime component.

Random Forest Feature Importance

Among the engineered features, the most recent lagged close (lag_1) dominated the Random Forest's predictions, accounting for approximately 82% of total feature importance, followed by the 20-day moving average (~15%). Shorter-horizon lags (lag_2, lag_3) and longer-horizon indicators (MA_50, Volatility_20, Volume, lag_5, lag_10) each contributed under 3% individually — reinforcing that yesterday's price is overwhelmingly the best single predictor of today's price in this market.

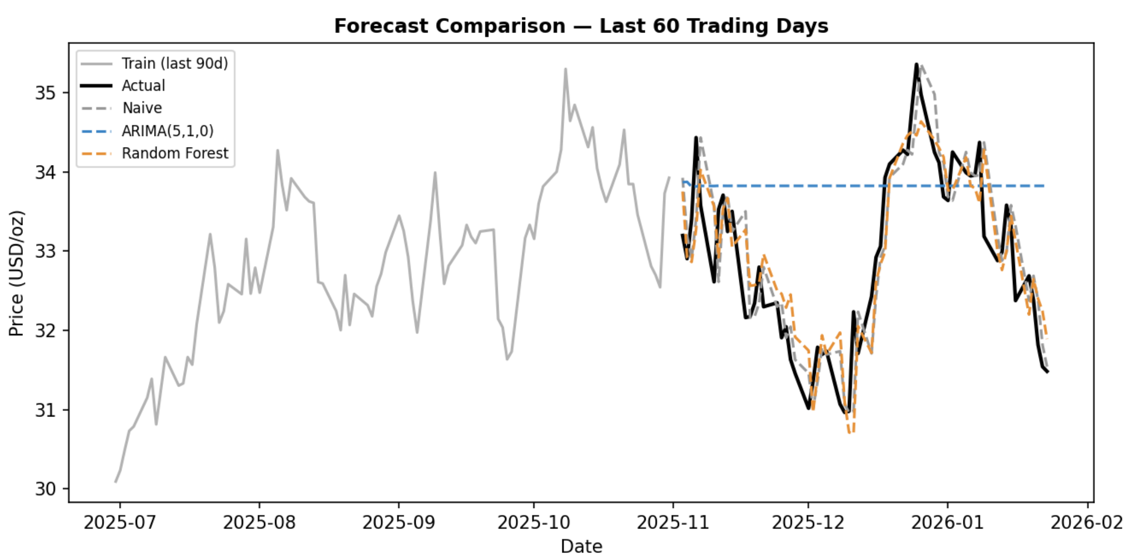

Figure 5. Actual vs. predicted close price for all three models over the 60-day test window.

6. 30-Day-Ahead Forecast

Re-fitting the ARIMA(5,1,0) model on the full 2,606-day history and projecting 30 trading days beyond the last observed date yields the forecast in Figure 6. Starting from the last actual close of $31.48 on January 23, 2026, the model projects a close of approximately $31.52 by March 6, 2026 — essentially unchanged — with a 95% confidence interval spanning $28.20 to $34.85.

Figure 6. 30-trading-day-ahead ARIMA forecast with 95% confidence interval.

The wide confidence band (roughly ±10% of the point forecast) is itself an important finding: it quantifies just how much uncertainty accumulates over a one-month horizon given silver's historical daily volatility of ~1.49%, and it should discourage over-reliance on the point forecast alone.

Disclaimer: *This forecast is an educational demonstration of time series methodology, not investment advice. Univariate price models cannot capture exogenous shocks (geopolitical events, central bank policy shifts, industrial demand shocks) that materially affect commodity prices.*

7. Conclusions & Recommendations

7.1 Summary of Findings

- Silver's decade-long price path is regime-driven, not trend-stationary: distinct multi-year bull and bear phases (2018 decline, 2019–2020 COVID-era surge, 2021–2023 consolidation, 2024–2025 rally) dominate the picture more than any single long-run trend.
- Daily returns are approximately symmetric with mild excess kurtosis — extreme single-day moves occur somewhat more often than a normal distribution would predict, but not dramatically so in this sample.

- Volatility is clearly regime-dependent and clusters in time, peaking sharply around the COVID-19 shock in mid-2020.
- Price levels are non-stationary; first differences are stationary — a standard result for asset prices that supports differencing-based models like ARIMA.
- Over a 60-day test horizon, none of the three models decisively beat a naive persistence forecast, underscoring how close daily silver prices are to a random walk at short horizons.

7.2 Recommended Next Steps

- Incorporate exogenous variables — gold price, US Dollar Index (DXY), real interest rates, and industrial demand proxies (e.g., solar panel installation data) — into a multivariate model (e.g., ARIMAX, VAR, or gradient-boosted trees with these features).
- Replace the single train/test split with walk-forward (rolling-origin) validation for a more robust, less split-dependent estimate of model accuracy.
- Explore deep learning approaches (LSTM/GRU) and Facebook Prophet as additional benchmarks, particularly for capturing longer seasonal cycles.
- Apply regime-detection methods (e.g., Hidden Markov Models or Markov-switching models) to formally identify and forecast transitions between the volatility regimes observed in Section 3.4.
- Extend the volatility analysis with a GARCH-family model, which is purpose-built for the volatility clustering documented in Section 3.4, rather than relying solely on a rolling standard deviation.

Appendix: Companion Materials

This report is accompanied by a fully executable Google Colab notebook (Silver_Market_Analysis_Forecasting.ipynb) that reproduces every statistic, table, and figure referenced above, including additional interactive visualizations (an OHLC candlestick chart and full seasonality boxplots) not reproduced here for brevity.